DO NOW · AO1 REVISION

Induced potential, transformers and the National Grid

Magnetism and electromagnetism
Paper 2 · Separate Physics only
22 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State what is induced by the generator effect.**	A potential difference (and a current, in a complete circuit).	☐
2 ★	**State what is needed for the generator effect to occur.**	A conductor moving relative to a magnetic field, or a changing magnetic field around a conductor.	☐
3 ★	**Give three ways to increase the size of the induced potential difference.**	Use a stronger magnetic field, move faster, or use more turns on the coil.	☐
4	**Describe how to reverse the direction of the induced potential difference.**	Reverse the direction of movement or of the magnetic field.	☐
5 ★	**State the difference between an alternator and a dynamo.**	An alternator generates alternating current; a dynamo generates direct current.	☐
6	**Name the effect used in microphones.**	The generator effect	☐
7 ★	**Describe how a microphone works.**	Sound makes a diaphragm and coil vibrate in a magnetic field, inducing a changing potential difference.	☐
8	**Name the effect used in transformers.**	The generator effect	☐
9 ★	**Describe the structure of a transformer.**	A primary coil and a secondary coil wound on an iron core.	☐
10	**Describe what a step-up transformer does.**	It increases the potential difference.	☐
11	**Describe what a step-down transformer does.**	It decreases the potential difference.	☐
12 ★	**Compare the numbers of turns on the coils of a step-up transformer.**	More turns on the secondary coil than on the primary.	☐

© 2026 PanPhy Labs. All rights reserved.

	QUESTION	ANSWER	✓
13	**Compare the numbers of turns on the coils of a step-down transformer.**	Fewer turns on the secondary coil than on the primary.	☐
14 ★	**State how power input and power output are related in an ideal transformer.**	They are equal.	☐
15	**Name the type of current supplied to the primary coil of a transformer.**	Alternating current	☐
16	**Name the type of current induced in the secondary coil of a transformer.**	Alternating current	☐
17 ★	**Explain why a transformer needs an iron core.**	Iron is easily magnetised, so it carries the changing magnetic field from the primary coil to the secondary coil.	☐
18 ★	**State what the magnetic field of an induced current does.**	It opposes the change that caused it.	☐
19 ★	**Explain why a transformer does not work with a direct current.**	Direct current gives a constant magnetic field; induction needs a changing field.	☐
20	**State the equation linking the potential differences and numbers of turns of a transformer.**	p.d. across primary ÷ p.d. across secondary = turns on primary ÷ turns on secondary ($V_p \div V_s = n_p \div n_s$)	☐
21 ★	**Describe the potential difference–time graph for an alternator.**	It alternates: a wave going equally above and below zero.	☐
22 ★	**Describe the potential difference–time graph for a dynamo.**	It rises and falls but never goes below zero.	☐

© 2026 PanPhy Labs. All rights reserved.